

ODD END SEMESTER EXAMINATION DECEMBER-2025

Roll no.

Name of the Course: B.Tech, CE (Specialization in RS & GIS)

Semester: III

Name of the Paper: Remote Sensing and Its Techniques
Time: 3 HourPaper Code: TCE 399
Maximum Marks: 100**Note:**

- (i) All questions are compulsory.
- (ii) Answer any two sub question among **a, b & c** in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1.	(20 Marks), CO1
a. Apply your knowledge to interpret the spectral response curve of vegetation, soil, and water.	
b. With examples, analyze how EMR interaction with atmosphere.	
c. Construct a labeled diagram showing the EMR spectrum relevant to optical sensors.	
Q2.	(20 Marks), CO2
a. Evaluate the comparative advantages of aerial photography in remote sensing.	
b. Design a stereoscopic setup and explain its working principle.	
c. Explain how a parallax bar can be used to compute object height from overlapping photos.	
Q3.	(20 Marks), CO3
a. Identify suitable remote-sensing (RS) platforms for mapping flood-prone areas.	
b. Distinguish between whiskbroom and pushbroom scanning techniques with neat sketches.	
c. Correlate reflectance values with digital number scaling in image preprocessing.	
Q4.	(20 Marks), CO4
a. Assess the contribution of weather/meteorological satellites to climate forecasting.	
b. Demonstrate how land-observation satellite data assist in urban growth detection.	
c. Compare specifications and uses of (any two) CARTOSAT-3, IKONOS, QUICKBIRD, and NOAA satellites.	
Q5.	(20 Marks), CO5
a. Explain the physical principle behind active and passive RS systems with diagrams.	
b. Critically analyze the benefits of microwave sensing for all-weather data acquisition.	
c. Prepare a flow diagram representing the complete process of image interpretation.	

